

# LAWRENCE FENG

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## Education

### Carnegie Mellon University

Fall 2022 – Spring 2026

*Bachelor of Science in Statistics and Machine Learning*

*Pittsburgh, Pennsylvania*

- Minor in Computer Science, minor in Mathematics. Admitted into 5th-year Masters in Machine Learning.
- GPA: 3.90

## Relevant Coursework

- Introduction to Deep Learning (PhD)
- Deep Reinforcement Learning (PhD)
- Probabilistic Graphical Models (PhD)
- Probability and Mathematical Statistics (PhD)
- Algorithm Design and Analysis
- Parallel and Sequential Data Structures
- Advanced Methods for Data Analysis
- Monte Carlo Methods and Applications

## Research Experience

### Honors Undergraduate Research Thesis

Fall 2025 – Present

*Advised by Professor Aditi Raghunathan*

- Led [Early Data Exposure Improves Robustness to Subsequent Fine-Tuning](#) — earlier versions of this work accepted to ICLR 2026 [SPOT, GRaM, DATA-FM]; under submission at COLM 2026
- Our work reframes catastrophic forgetting as a property of upstream training rather than a downstream mitigation problem; *how* a model acquires a capability matters for how it is retained, revealing false equivalences in prior work
- We study what we call early exposure, that is mixing target data into pretraining, revealing performance gains on that target domain *after* a downstream fine-tuning stage is introduced; early exposure also improves Pareto frontiers
- Developed theory showing that specialized features are protected from downstream interference using early exposure

### Carnegie Mellon Computer Science Department

Fall 2024 – Summer 2025

*Advised by Professor Wenting Zheng*

- [Developed novel watermark embedding algorithm which makes cryptographic watermarking \(more\) practical](#)
- Sole programmer managing all aspects: experimental design, implementation, analysis, and progress reporting
- Synthesizing complex experimental results into clear visualizations and accessible summaries for research collaboration

### Robotics Institute at Carnegie Mellon

Spring 2024 – Fall 2024

*Summer Research Apprenticeship*

- Explored adapting CoTracker's retrospective dense point tracking into predictive models that output future robot states
- Uncovered significant robustness failures in state-of-the-art models through systematic environmental variations

## Work and Leadership Experience

### Inflection AI

Summer 2025

*Member of Technical Staff Intern*

- Developed post-training tools for the open-source torchtune library, including support for distributed [Qwen2.5-VL fine-tuning \(PR #2868\)](#); required deep understanding of multimodal RoPE and fusion architectures
- Proposed and built benchmark refinement tool leveraging Item Response Theory to transform statistically insignificant LLM-as-a-judge evaluations into reliable discriminative assessments, now part of the company's core evaluation suite
- Core contributor to a company-wide effort on EQ evaluation; surveyed prior work and collaborated on designing a suite of benchmarks, prototyped persona-based LLM simulators, and built synthetic data generation pipelines

### Carnegie AI Safety Initiative

Fall 2023 – Present

*President (Fall 2025 - Spring 2026), Member*

- Spearheaded outreach growing mailing list 4× to ~ 900, expanded leadership team 4× to ~ 16, developed and taught [technical AI safety fundamentals course](#) to ~ 40 ppl/sem, piloted and grew jailbreaking hackathon to ~ 200 participants

## Projects

### Interpreting Vision Language Models | *Python, Hugging Face, TransformerLens*

Fall 2024

- Found that a language-only sparse autoencoder can provide insights into a multimodal model's black-box behavior
- Demonstrated the ability to manipulate how a model interprets images using natural language concepts

## Technical Skills

**Programming Languages:** Python, R, C, C++, Java, SQL, SML,  $\LaTeX$ , Bash

**Developer Tools:** GCP, AWS EC2, Git, WandB, Slurm, Conda, uv

**Frameworks and Libraries:** PyTorch, TensorFlow, Hugging Face, vLLM, OpenCV, NumPy, Pandas, Matplotlib, ROS